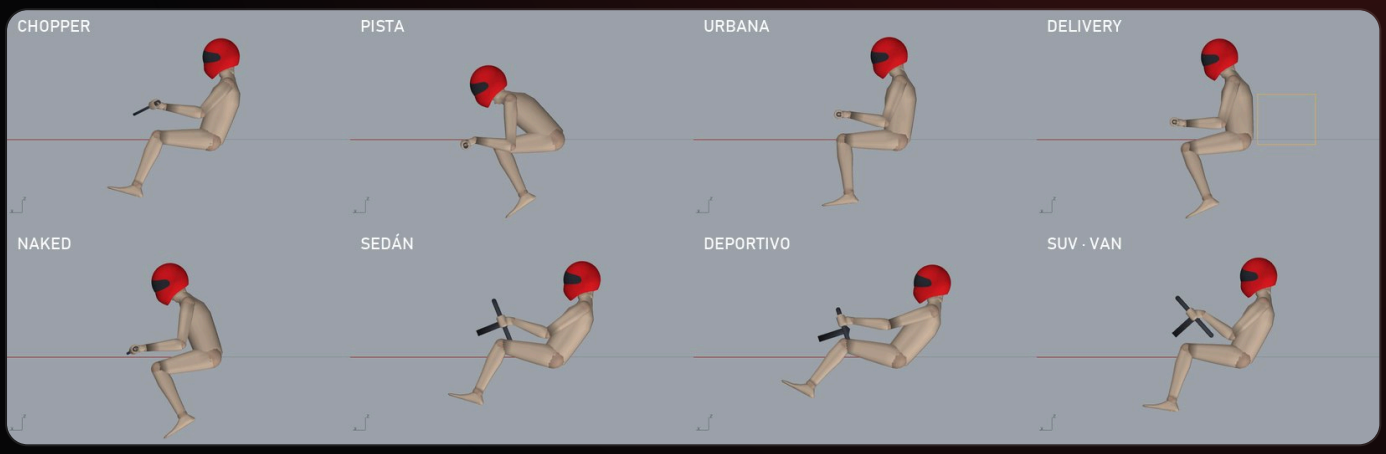
# JAGUAR for Rhino

HUMAN FACTORS · KINEMATICS · VEHICLE ERGONOMICS

F3DE Design

USER MANUAL · V1.0.0

A free plugin for **Rhino 8** that inserts a **virtual biped (mannequin)** into your model to design and compare the ergonomics and packaging of **cars and motorcycles**, based on the SAE J826 (H-point) mannequin, Dreyfuss (1966) and ANSUR II, with no Grasshopper required.




Eight preset postures: five for motorcycles (chopper, track, urban, delivery, naked) and three for cars (sedan, sport, SUV).

## HOW IT WORKS

Type the **Jaguar** command or click the jaguar icon in the toolbar. A panel opens with collapsible groups, in this order:

### 1 *Profile*

Pick the person you're modeling: **world region** (18, each with its own average stature), **male or female**, and **percentile from P1 to P99**. Body measurements (stature, thigh, leg and foot) update instantly and the whole mannequin scales.

### 2 *Postures*

Switch between **motorcycle** and **car** and pick a preset posture:

- **Motorcycle:** chopper, track, urban, delivery and naked, with different handlebar types (straight, touring, cruiser, clip-on) and the standing posture.
- **Car:** sedan, sport and SUV, with a **hand clock** (from nine-and-three to eleven-and-one) to place the hands on the wheel.
- **Wheel / handlebar switch:** on, the hands always grip the wheel or the handlebar; off, everything moves freely.
- **Adjustable steering wheel:** distance, height, rake and diameter.




JAGUAR panel: Profile and Postures.




Occupants group.

KEY FEATURES

### 3 Angles · Fine tuning

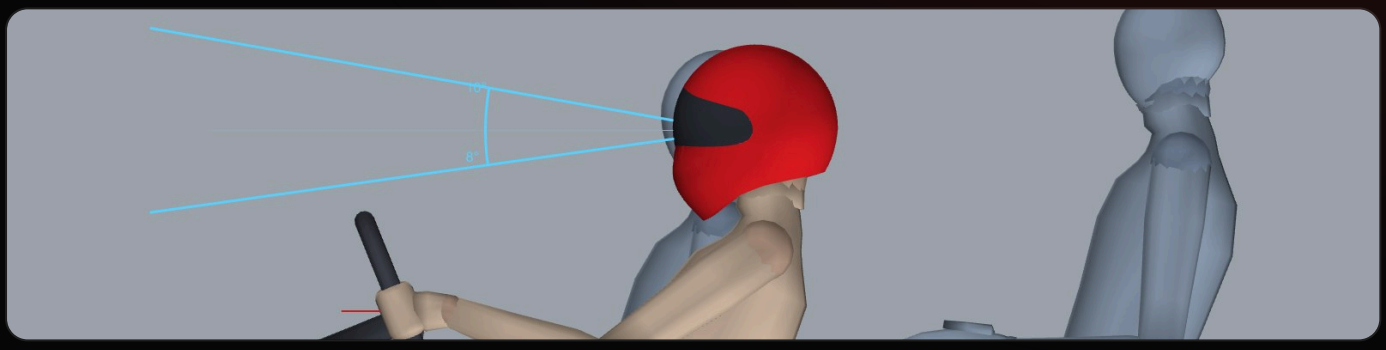
Every joint can be corrected by hand: spine, neck, hip (femur), knees, shin, foot, shoulders, elbows and wrists. The hands keep following the handlebar or wheel even as you move everything else.

### 4 Occupants

Size the cabin with more than one person: a **co-driver** (left or right), **up to 6 rear passengers** with adjustable seat height, and on motorcycles a **helmeted pillion** with different arm positions (waist, shoulders, seat or free). Each group has its own sex and percentile.

### 5 Ergonomics

Shows hip, knee, ankle and elbow angles against their **comfort ranges**, and the key distances from the H-point.



Driver and rear passenger with the vision angles toward the windshield (10° up and 8° down, both editable).

### 6 Loads and fatigue

Turn on the **colour map** over the body: from **blue** (low load) to **red** (high load). It computes static joint moments and factors in **time** in the posture, **vibration** on the seat and the handlebar, **speed** (on motorcycles) and **acceleration or braking** (in g). Useful to **compare postures** and spot the most demanding areas.

low load — high load

### 7 Visibility

Draws the **vision angles** from the eyes, up and down, to size the windshield.

### 8 View

Shows or hides the **biped** and the **skeleton** independently.




Loads and fatigue group.



## WORKFLOW

1. **Pick the profile**
   Region, sex and percentile of who will use the vehicle.
2. **Choose a posture**
   Car or motorcycle, then adjust the wheel or the handlebar.
3. **Fine-tune the angles**
   Correct the joints and add occupants.
4. **Analyze**
   Check ergonomics, loads and visibility.
5. **Bake in Rhino**
   The model is virtual and is only created in your scene when you click "Bake in Rhino". "Reset" goes back to the start.

## INSTALLATION

1. Download **JAGUAR-Installer.zip** and unzip it.
2. **Close every Rhino window** and run **JAGUAR-Setup.exe** (if your antivirus blocks it, use **INSTALAR.cmd**).
3. Open Rhino 8: you'll see the JAGUAR toolbar. If it doesn't show up, type the **Jaguar** command.

Requirements: Windows 10/11 and Rhino 8. Panel language: Spanish and English (ES / EN button).

## LINKS

| | |
|---|---|
| DOWNLOAD | github.com/fedemergent/jaguar-rhino/releases/latest |
| VIDEO TUTORIAL | youtu.be/F03E_xMneso |
| README ON GITHUB | github.com/fedemergent/jaguar-rhino |
| F3DE DESIGN | fedemergent.github.io/f3de-design |

## LICENSE AND SCOPE

JAGUAR is free for **non-commercial** use, under the **CC BY-NC-SA 4.0** license: attribution to the author (Federico Escobar Salas · fedemergent · F3DE Design), no commercial use, and share-alike.

It is a reference tool, not a certification. Anthropometric data and the load analysis are **indicative** (a static sagittal-plane screening) and are meant to compare postures; they do not replace a certified ergonomic study.

## STANDARDS AND UNDERLYING DATA

- SAE J826: H-point mannequin.
- SAE J1100, J941 and J1050: package, eye ellipse and field of view.
- Dreyfuss (1966) and ANSUR II: body segments and measurements.
- Dempster / Winter, Rohmert and Directive 2002/44/EC: masses, endurance and vibration.

## TRADEMARK NOTICE

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